What Societies Do, Not What They Believe: Behavioral Cultural Practices as Predictors of COVID-19 Mortality

Pat Brenner

Abstract

Background. Cultural dimensions influence how societies perceive, respond to, and recover from collective threats. Yet their role in shaping pandemic outcomes has been largely overlooked in global public health modeling. This study investigates associations between national cultural values and COVID-19 mortality to generate new hypotheses for integrative health forecasting.

Methods. We combined COVID-19 mortality data from late 2021 with cross-national cultural measures from the GLOBE Project's nine societal practice dimensions and Hofstede's six cultural dimensions, along with demographic controls. Staged regressions compared each framework's incremental predictive contribution above demographics. Interpretive analysis explored mechanisms through which significant predictors may influence pandemic behavior and outcomes.

Findings. GLOBE societal practices dramatically outperformed both demographics and Hofstede's framework. Adding GLOBE practices to demographic controls increased adjusted R^2 from 0.245 to 0.512 (N = 49). By contrast, Hofstede's dimensions added negligibly to demographics (adjusted R^2 from 0.091 to 0.097). Gender Egalitarianism emerged as the most robust cultural predictor of COVID-19 mortality, with a significant positive association across all model specifications. Humane Orientation showed a protective effect. Institutional Collectivism exhibited the strongest bivariate correlation with mortality but shared variance with demographic controls. Health expenditure per capita showed effectively zero association with outcomes ($r = 0.018$). GLOBE societal practices outperformed their corresponding values dimensions in seven of nine comparisons, indicating that what societies actually do predicts pandemic outcomes far better than what they aspire to.

Conclusions. This study highlights the need to integrate cultural analytics into global health policy, surveillance, and response frameworks. The finding that gender-egalitarian societies

experienced significantly higher mortality—not because equality is harmful, but because the social structures accompanying it create wider transmission surfaces—carries direct implications for pandemic preparedness in modernized democracies. Cultural traits governing authority, cooperation, and gender dynamics are not peripheral to pandemic outcomes but central to understanding national variation in mortality. Resource investment without cultural alignment produces poor returns. Future pandemic planning should incorporate culturally calibrated modeling, messaging, and institutional design.

1. Introduction

This study is intended as a hypothesis-generating analysis, offering preliminary insights into how cultural dimensions may correlate with COVID-19 mortality across countries. Rather than establishing definitive causal relationships, we aim to illuminate underexplored sociocultural patterns that warrant further investigation through longitudinal and mixed-method research.

Despite increasing attention to the social determinants of health, cultural dimensions have remained relatively underexplored in global pandemic policy frameworks. Many public health models emphasize biomedical or infrastructural variables, with culture often treated as a background factor rather than a predictive one. This omission may reflect both the methodological challenges of quantifying culture and the historic dominance of universalist approaches in epidemiology.

The global variation in COVID-19 mortality rates cannot be fully explained by medical capacity or public health interventions alone. Cultural norms and demographic structures shape individual behavior, risk perception, and social cohesion—all of which influence the transmission and lethality of pandemics. While many studies have examined biomedical and political factors associated with COVID-19 outcomes, fewer have considered the embedded social dimensions that condition how populations respond to crisis. Culture influences adherence to health directives, interpersonal distancing, information trust, and emotional coping—all of which impact pandemic spread.

This study tests the predictive power of cultural dimensions using two prominent global frameworks: the GLOBE Project's societal practice dimensions, which measure what societies actually do, and Hofstede's cultural dimensions, which capture more abstract value orientations. Recent work by Dheer et al. (2021) has demonstrated significant cross-national associations between cultural values and pandemic outcomes, reinforcing the relevance of this line of inquiry. We examine their contributions relative to traditional demographic indicators and propose an integrated model of COVID-19 vulnerability that bridges cultural psychology and global public health. Our aim is to generate empirical clarity around how different dimensions of national character may buffer or exacerbate mortality in times of global crisis.

A central question animating this analysis is whether behavioral cultural practices or aspirational cultural values better predict real-world health outcomes. The GLOBE Project uniquely distinguishes between "as is" practices and "should be" values for each of its nine dimensions, enabling a direct test of this question. If practices consistently outperform values, it suggests that pandemic preparedness models should attend to what societies do rather than what they believe.

2. Methods

We compiled a cross-sectional dataset of 94 countries using publicly available data sources. COVID-19 mortality was measured as cumulative deaths per 1,000 people as of December 31, 2021, obtained from Our World in Data. This cutoff was chosen to capture the bulk of pandemic-related deaths while minimizing distortions from subsequent mutations or reporting gaps.

China was excluded from the analysis due to widely documented concerns about the reliability of its official COVID-19 mortality data. China's reported mortality of 2.8 deaths per million is orders of magnitude below estimates from excess mortality analyses and epidemiological models. Section 5.6 presents quantitative evidence from our model that independently corroborates this assessment. The exclusion of the world's most populous nation and one of the richest cultural datasets in the GLOBE framework is regrettable but methodologically necessary; including demonstrably unreliable outcome data would compromise the integrity of the analysis. Future research using validated excess mortality estimates could potentially reincorporate China.

2.1 Variables

Dependent variable. COVID-19 cumulative deaths per 1,000 population as of December 31, 2021 (Our World in Data).

Demographic controls. Three non-redundant demographic variables were selected to minimize multicollinearity while capturing key structural differences: median age (UN Population Division), Gini coefficient of income inequality (World Bank), and population density in persons per square mile (derived from UN and national census data). An initial screening of 15 candidate demographic variables—including age cohort percentages, health expenditure per capita, hospital capacity, urbanization, air quality indices, obesity prevalence, and multiple fractionalization measures—revealed extensive redundancy and negligible bivariate associations

with mortality for most candidates. The trimmed set of three controls retained the most theoretically relevant and statistically independent measures.

Cultural variables. Cultural dimensions were drawn from two frameworks:

The GLOBE Project's nine societal practice dimensions (House et al., 2004): Uncertainty Avoidance, Future Orientation, Power Distance, Institutional Collectivism, Humane Orientation, Performance Orientation, In-Group Collectivism, Gender Egalitarianism, and Assertiveness. GLOBE uniquely distinguishes between societal practices ("as is") and societal values ("should be"); we used practices as primary predictors and values for comparative analysis.

Hofstede's six cultural dimensions (Hofstede, 2001): Power Distance, Individualism–Collectivism, Masculinity–Femininity, Uncertainty Avoidance, Long-Term Orientation, and Indulgence–Restraint.

2.2 Analytical Approach

Analysis proceeded in a staged design with matched sample sizes to enable fair comparison across frameworks:

Stage 1: Demographic baselines. Ordinary Least Squares (OLS) regression of deaths per 1,000 on the three demographic controls, run separately at each cultural framework's available sample size (N = 49 for GLOBE countries, N = 55 for Hofstede countries). This provides the correct baseline for evaluating each framework's incremental contribution.

Stage 2: Demographics plus culture. OLS regression adding each cultural framework's full set of dimensions to the demographic controls. The primary model is Demographics + GLOBE Practices (N = 49). The comparison model is Demographics + Hofstede (N = 55).

Stage 3: GLOBE practices only. OLS regression using GLOBE practices without demographic controls, to assess whether culture alone explains mortality variation and whether demographics add meaningful incremental power.

Additionally, bivariate Pearson correlations were computed between all candidate predictor variables and COVID-19 deaths per 1,000, serving as a preliminary screening step. Multicollinearity was evaluated using Variance Inflation Factors (VIF), with VIF > 10 flagged

for interpretation caution. All statistical analyses were conducted using Python with NumPy and SciPy.

3. Results

3.1 Bivariate Correlations

Table 1 presents the bivariate correlations between all candidate predictors and COVID-19 deaths per 1,000. Six of the eight strongest correlates are GLOBE societal practice dimensions, with Institutional Collectivism ($r = -0.525$, $p < 0.001$), Humane Orientation ($r = -0.480$, $p < 0.001$), and Gender Egalitarianism ($r = 0.422$, $p = 0.002$) occupying the top three positions. Hofstede's Uncertainty Avoidance ranked sixth ($r = 0.337$, $p = 0.006$), the only Hofstede dimension with a statistically significant bivariate association. Median age ranked fourth ($r = 0.359$, $p < 0.001$), outperformed by three cultural dimensions. Health expenditure per capita showed effectively zero association with mortality ($r = 0.018$, $p = 0.864$, $N = 89$).

Table 1

Bivariate correlations between candidate predictors and COVID-19 deaths per 1,000.

Rank	Source	Variable	R	r^2	p	N
1	GLOBE Practice	Institutional Collectivism	−0.525	0.276	<0.001	50
2	GLOBE Practice	Humane Orientation	−0.480	0.231	<0.001	50
3	GLOBE Practice	Gender Egalitarianism	0.422	0.178	0.002	50
4	Demographic	Median Age	0.359	0.129	<0.001	92
5	GLOBE Practice	Future Orientation	−0.358	0.128	0.011	50
6	Hofstede	Uncertainty Avoidance	0.337	0.114	0.006	65
7	GLOBE Practice	Performance Orientation	−0.333	0.111	0.018	50
8	GLOBE Value	Gender Egalitarianism	0.311	0.097	0.028	50
9	GLOBE Practice	Uncertainty Avoidance	−0.286	0.082	0.044	50
…						
20	Demographic	Pop. Density	−0.162	0.026	0.121	93
—	Demographic	HP Spend per Capita	0.018	<0.001	0.864	89

Note. Ranks shown for top predictors and selected non-significant variables.

3.2 Practices Versus Values

A systematic comparison of GLOBE practice and value dimensions revealed that practices outperformed their corresponding values in seven of nine dimension pairs (Table 2). The most dramatic divergence occurred in Institutional Collectivism, where the practice dimension ranked first overall (r = −0.525) while its value counterpart showed essentially zero association (r = 0.008, rank 27). This pattern held across the most consequential dimensions: Humane Orientation practices (r = −0.480) versus values (r = −0.118); Performance Orientation practices (r = −0.333) versus values (r = −0.149); and Uncertainty Avoidance practices (r = −0.286) versus values (r = −0.107). These findings indicate that what societies actually do—their behavioral norms, institutional routines, and enforcement patterns—predicts pandemic outcomes far more reliably than what they aspire to.

Table 2

Bivariate correlations with COVID-19 deaths per 1,000

Dimension	Practice r	Practice p	Value r	Value p	Practice Wins?
Institutional Collectivism	−0.525	<0.001	0.008	0.958	Yes
Humane Orientation	−0.480	<0.001	−0.118	0.413	Yes
Gender Egalitarianism	0.422	0.002	0.311	0.028	Yes
Future Orientation	−0.358	0.011	−0.240	0.093	Yes
Performance Orientation	−0.333	0.018	−0.149	0.301	Yes
Uncertainty Avoidance	−0.286	0.044	−0.107	0.461	Yes
Assertiveness	0.170	0.238	−0.169	0.240	Yes
In-Group Collectivism	−0.093	0.522	−0.152	0.291	No
Power Distance	−0.010	0.945	−0.108	0.458	No

Note. GLOBE practices versus values for each dimension pair.

3.3 Demographics-Only Baseline

The demographics-only model at the GLOBE sample size (N = 49) explained 24.5% of the variance in COVID-19 deaths per 1,000 (adjusted $R^2 = 0.245$, $F(3,45) = 6.18$, $p = 0.001$). All three demographic controls reached statistical significance: median age ($\beta = 0.077$, $p < 0.001$), Gini coefficient ($\beta = 0.068$, $p = 0.004$), and population density ($\beta = -0.0001$, $p = 0.024$). Older populations, more unequal societies, and less dense countries experienced higher mortality.

At the Hofstede sample size (N = 55), demographics explained less variance (adjusted $R^2 = 0.091$, $F(3,51) = 2.80$, $p = 0.049$), reflecting differences in the country composition of the two subsamples.

3.4 Primary Model: Demographics + GLOBE Practices

The primary model (N = 49) explained 51.2% of the variance in COVID-19 deaths per 1,000 (adjusted $R^2 = 0.512$, $F(12,36) = 5.20$, $p < 0.0001$). Adding GLOBE practices to the demographic baseline increased adjusted R^2 by 0.268 percentage points—roughly doubling the model's explanatory power. Table 3 presents the full regression results.

Table 3

Primary regression model: Demographics + GLOBE Practices

Variable	B	SE	β	t	p	VIF	Sig.
(Intercept)	−3.328	4.578	—	−0.73	0.472	—	
Median Age	0.033	0.022	0.317	1.55	0.130	4.11	
Gini Coefficient	0.044	0.021	0.365	2.12	0.041	2.93	*
Population Density	<−0.001	<0.001	−0.454	−2.89	0.007	2.43	**
GLOBE Uncertainty Avoid.	−0.026	0.336	−0.019	−0.08	0.938	5.85	
GLOBE Future Orient.	0.133	0.359	0.076	0.37	0.713	4.16	
GLOBE Power Distance	0.008	0.389	0.004	0.02	0.983	3.31	
GLOBE Inst. Collectivism	−0.462	0.373	−0.233	−1.24	0.224	3.49	

Variable	B	SE	β	t	p	VIF	Sig.
GLOBE Humane Orient.	−0.801	0.348	−0.466	−2.30	0.027	4.04	*
GLOBE Perform. Orient.	0.410	0.368	0.194	1.11	0.273	3.00	
GLOBE In-Group Collect.	0.268	0.240	0.240	1.12	0.272	4.56	
GLOBE Gender Egalitar.	1.103	0.325	0.456	3.39	0.002	1.78	**
GLOBE Assertiveness	−0.076	0.342	−0.032	−0.22	0.825	2.07	

Note: *N = 49. Adjusted R^2 = 0.512. B = unstandardized coefficient; β = standardized coefficient.*

**p < 0.05, **p < 0.01.*

Two cultural predictors reached statistical significance. Gender Egalitarianism showed a strong positive association with mortality (β = 1.103, p = 0.002), meaning more gender-egalitarian societies experienced significantly higher COVID-19 death rates. Humane Orientation showed a protective negative association (β = −0.801, p = 0.027), meaning more prosocial societies experienced lower mortality. Institutional Collectivism, despite having the strongest bivariate correlation (r = −0.525), did not reach significance in the full model (β = −0.462, p = 0.224), reflecting shared variance with other predictors. Among demographic controls, population density and Gini coefficient were significant; median age, despite its strong bivariate correlation, lost significance when cultural dimensions were included.

All VIF values remained below 6, indicating acceptable multicollinearity levels.

3.5 Comparison Model: Demographics + Hofstede

The Hofstede model (N = 55) explained only 9.7% of variance (adjusted R^2 = 0.097), and the overall model did not reach statistical significance ($F(9,45) = 1.65$, p = 0.131). Only Uncertainty Avoidance reached significance (β = 0.018, p = 0.048). The five remaining Hofstede dimensions showed no significant association with mortality. Adding Hofstede's six dimensions to demographics improved adjusted R^2 by just 0.006—essentially no improvement—compared to the 0.268 improvement from adding GLOBE practices.

3.6 Incremental Contributions Summary

Table 4

Model comparison summary

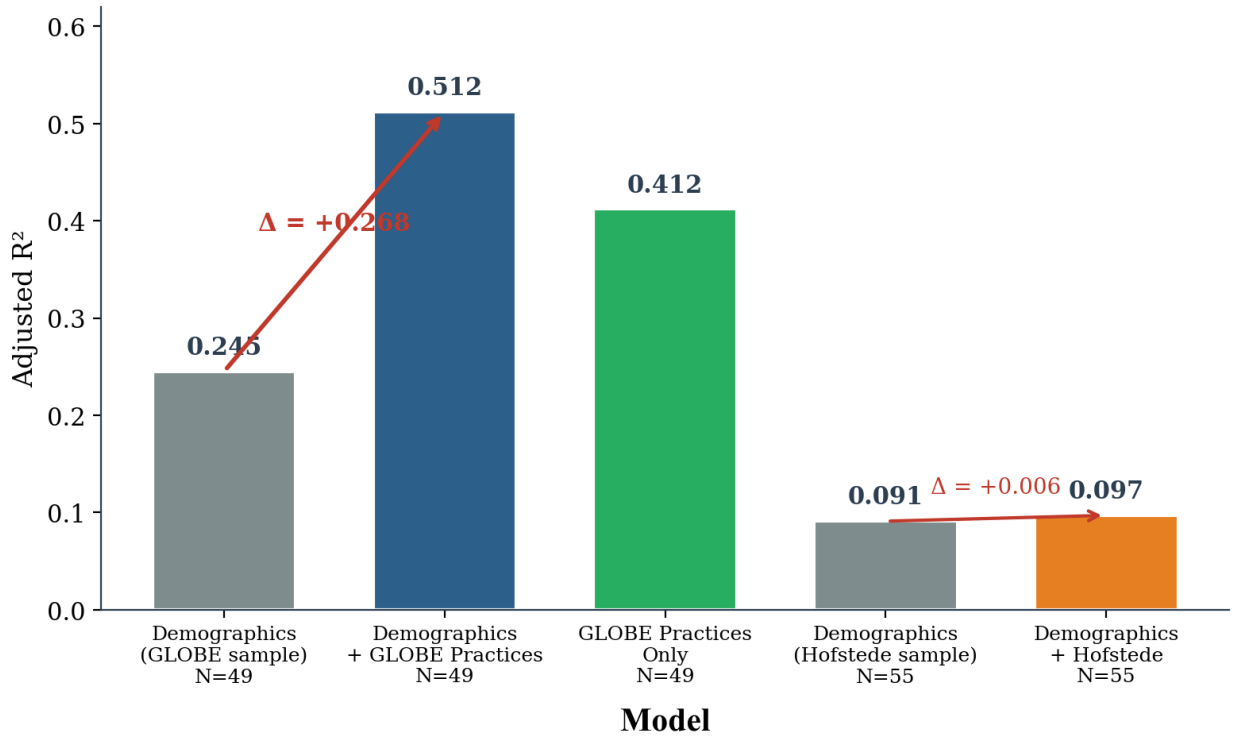
Model	N	R^2	Adj. R^2	F	p(F)
Demographics (GLOBE sample)	49	0.292	0.245	6.18	0.001
Demographics + GLOBE [Primary]	**49**	**0.634**	**0.512**	**5.20**	**<0.001**
GLOBE only (no demographics)	49	0.522	0.412	4.73	<0.001
Demographics (Hofstede sample)	55	0.142	0.091	2.80	0.049
Demographics + Hofstede	55	0.248	0.097	1.65	0.131

Note: The primary model (highlighted) explains over five times the variance of the Hofstede comparison model.

The incremental contribution analysis reveals that GLOBE practices are the dominant source of explanatory power. Adding GLOBE practices to demographics increases adjusted R^2 by +0.268 (from 0.245 to 0.512). Adding demographics to GLOBE practices increases adjusted R^2 by only +0.101 (from 0.412 to 0.512). By contrast, adding Hofstede to demographics increases adjusted R^2 by merely +0.006. Culture—specifically, GLOBE's behavioral practice measures—is doing most of the explanatory work.

Figure 1

Framework comparison between demographics, GLOBE, and Hofstede dimensions.



Note. Including Demographics and GLOBE Practices increased R^2 by 0.268 over Demographics alone, and by 0.100 over GLOBE Practices alone. Including Hofstede with Demographics increased R^2 by only 0.006.

4. Discussion

Our findings underscore the importance of cultural context in shaping pandemic vulnerability. While demographic structure—especially age—remains a meaningful predictor of COVID-19 mortality, cultural dimensions—especially those associated with gender norms, prosociality, and cooperation—convey substantial additional explanatory power. The primary model explains over half the cross-national variance in COVID-19 mortality, with cultural practices contributing the majority of that explanatory power.

The three significant or near-significant cultural predictors in the primary model tell a coherent story about the social ecology of pandemic vulnerability. Gender Egalitarianism describes the exposure surface—the structural conditions that determine how many contact points a society generates in daily life. Humane Orientation describes the voluntary behavioral response—the prosocial instincts that generate organic compliance with protective measures. Institutional Collectivism, though weakened in the multivariate model, describes the structural response capacity—the degree to which societies can mount coordinated institutional action. Together, these dimensions suggest that pandemic outcomes are shaped by the interplay of exposure, voluntary restraint, and institutional coordination.

4.1 Gender Egalitarianism: The Paradox of Progress

Gender Egalitarianism was the most robust predictor of COVID-19 mortality in this study, reaching significance across every model specification ($p = 0.002$ in the primary model). Its positive coefficient means that more gender-egalitarian societies experienced significantly higher death rates. This finding is counterintuitive and potentially controversial, but the underlying mechanism is structural rather than normative.

Gender-egalitarian societies are characterized by high rates of dual-income households, broad workforce participation across genders, independent living arrangements, and shared caregiving responsibilities (World Economic Forum, 2020). These features of social organization—which are generally desirable from equity and development perspectives—create a wider transmission surface during a pandemic that requires physical isolation. Compared to societies where large portions of the population are home-based by default, egalitarian societies generate more daily

interpersonal contact points through commuting, workplace interaction, childcare logistics, and distributed social networks.

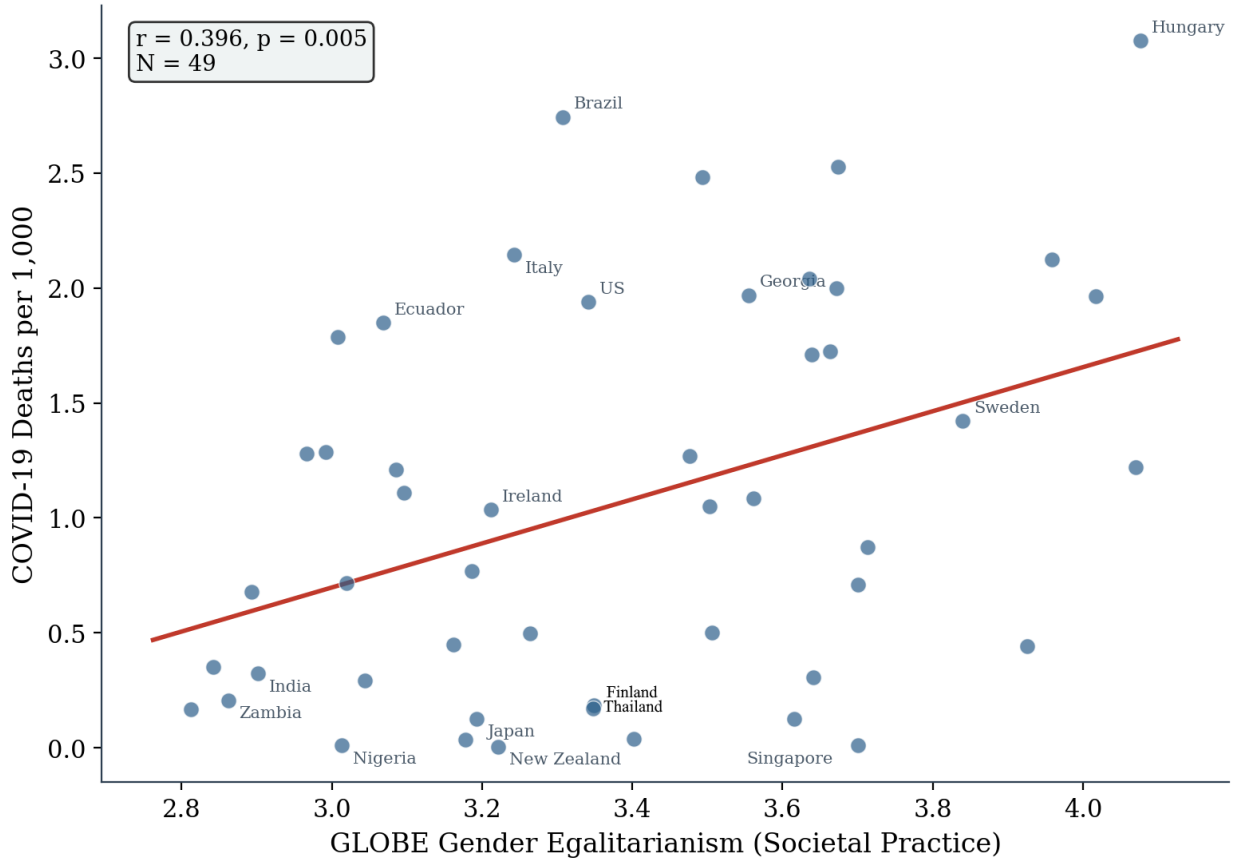
Additionally, gender-egalitarian societies tend to emphasize personal autonomy over social conformity (Inglehart & Norris, 2003), which may slow the adoption of collective mitigation behaviors such as masking, voluntary isolation, and compliance with mobility restrictions. The same cultural values that promote individual agency in normal times can become liabilities during a crisis that demands coordinated restraint. This interpretation aligns with Gelfand et al. (2021), who found that culturally "loose" societies—which tend to overlap with egalitarian ones—showed weaker early adoption of collective mitigation behaviors. Navarro et al. (2021) further documented that societies prioritizing economic continuity and individual productivity over collective restraint experienced higher excess mortality, suggesting a behavioral pathway linking egalitarian social organization to pandemic vulnerability.

Critically, this finding is not an argument against gender equality. It is an argument that the social architecture of egalitarian societies creates a specific pandemic vulnerability profile that must be accounted for in preparedness planning. A gender-egalitarian society is not more vulnerable because equality is harmful—it is more vulnerable because its social fabric is organized differently, generating different risk surfaces. The policy implication is not to reverse progress but to design pandemic responses that are calibrated to the mobility and autonomy patterns of modernized, egalitarian societies.

The fact that Gender Egalitarianism outperformed every other predictor—including age, wealth, and health spending—in robustness and consistency across model specifications suggests it captures something fundamental about how social modernization reshapes disease transmission dynamics. This finding warrants dedicated investigation in future pandemic preparedness research.

Figure 2

Bivariate relationship between GLOBE Gender Egalitarianism (societal practice) and COVID-19 deaths per 1,000



Gender Egalitarianism and COVID-19 Mortality

Note. Higher gender egalitarianism is associated with higher mortality (N = 49, r = 0.396, p = 0.005).

4.2 Humane Orientation: The Prosocial Brake

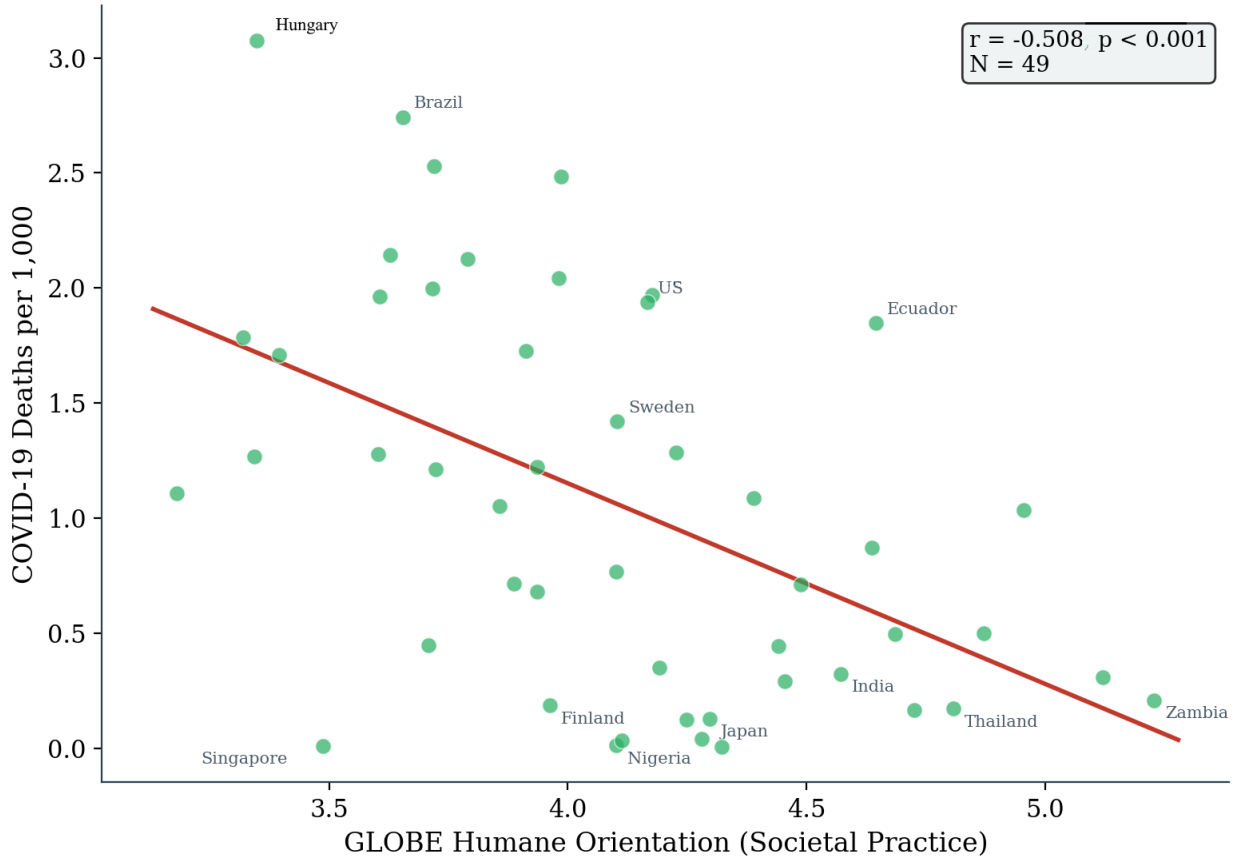
Humane Orientation showed a protective association with mortality (β = −0.801, p = 0.027), meaning societies that encourage compassion, altruism, and interpersonal concern experienced lower COVID-19 death rates. This dimension reflects societal norms around kindness, generosity, and obligation to others (House et al., 2004).

The protective mechanism likely operates through voluntary compliance. In societies where prosocial norms are strong, individuals are more likely to adopt protective behaviors—masking, distancing, self-isolation—not because they are mandated but because they feel responsibility toward their community. Research has linked prosociality with stronger community response during crises (Taylor et al., 2008). Countries scoring high on Humane Orientation may also maintain stronger informal caregiving networks and norms that support protecting the vulnerable. Shi et al. (2024) found that communal norms in tight Asian cultures predicted stronger vaccine intentions, providing additional support for the prosocial compliance pathway. Ma et al. (2024) showed that cultural tightness enhanced disease prevention vigilance even in high-vaccination contexts, suggesting that prosocial behavioral norms operate independently of medical infrastructure.

When considered alongside Gender Egalitarianism, Humane Orientation suggests a potential buffering mechanism. Egalitarian societies with strong prosocial norms may partially offset their structural vulnerability through organic behavioral restraint. The societies most at risk, by this logic, would be those with high gender egalitarianism but low prosocial orientation—high mobility and autonomy without the communal instinct to self-regulate during crisis.

Figure 3

Bivariate relationship between GLOBE Humane Orientation (societal practice) and COVID-19 deaths per 1,000



Humane Orientation and COVID-19 Mortality

Note. Higher prosocial orientation is associated with lower mortality ($N = 49$, $r = -0.508$, $p < 0.001$).

4.3 Institutional Collectivism

Institutional Collectivism exhibited the strongest bivariate correlation with COVID-19 mortality ($r = -0.525$, $p < 0.001$), consistent with prior findings that collectivist cultures comply more readily with public health mandates (Markus & Kitayama, 1991; Lu et al., 2021). Societies high in institutional collectivism prioritize group goals, social harmony, and norm enforcement—qualities conducive to coordinated responses and adherence to restrictions (Biddlestone et al., 2020). This dimension aligns conceptually with Gelfand's cultural tightness–looseness

continuum, where tighter cultures imposed stricter and more effective containment strategies (Gelfand et al., 2021). Han et al. (2021) found that authoritarian and communitarian cultures implemented and adhered to mitigation strategies more effectively than liberal-individualist counterparts. Kawamura et al. (2023) showed that cultural tightness was associated with higher voluntary vaccination rates among students in Japan, India, and the U.S., reinforcing the link between collective orientation and coordinated health behavior.

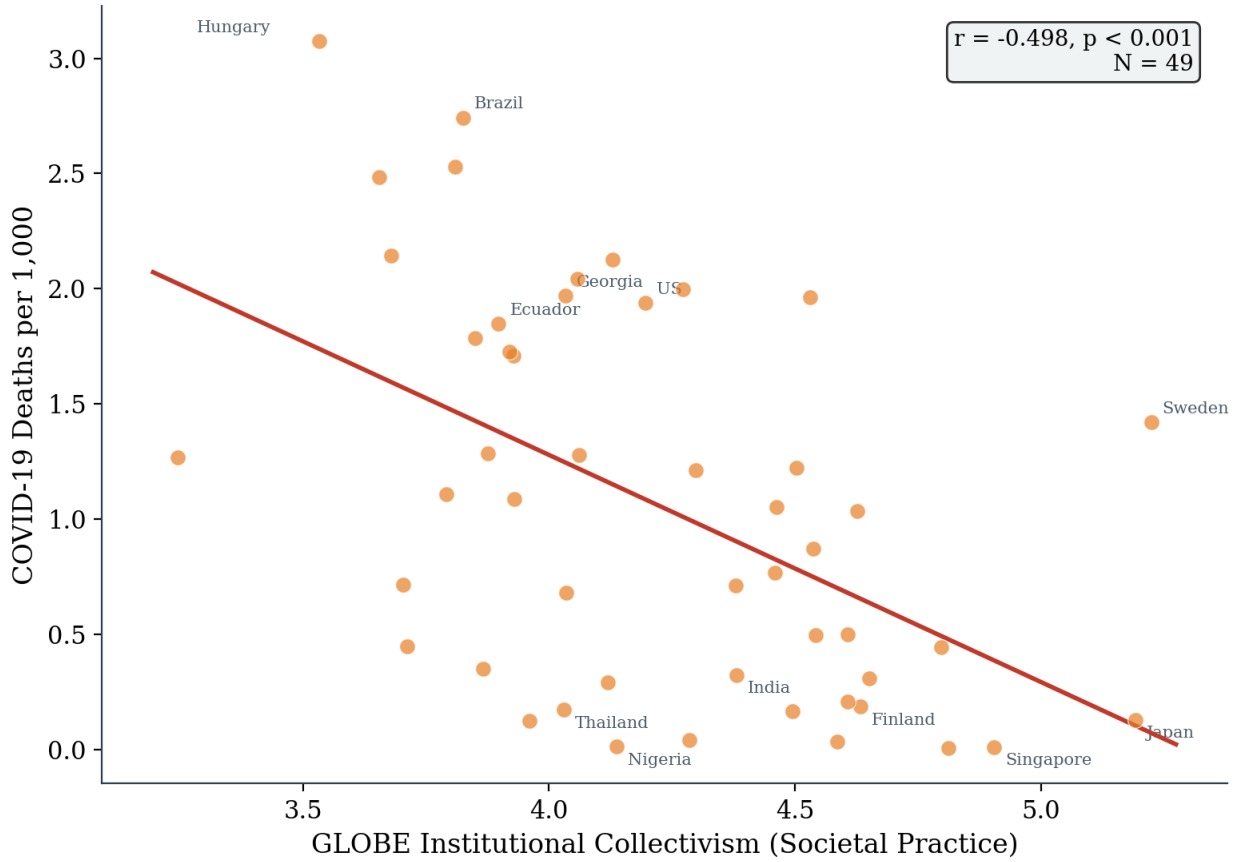
However, Institutional Collectivism did not reach significance in the primary multivariate model ($p = 0.224$). This weakening reflects the complex correlation structure among predictors. Humane Orientation correlates substantially with both median age ($r = -0.52$) and Institutional Collectivism ($r = 0.47$). When demographic controls are introduced, they absorb variance that was being attributed to Humane Orientation, which in turn affects the shared variance between Humane Orientation and Institutional Collectivism. The result is an indirect redistribution of explanatory credit, not evidence that collectivism is unimportant.

Schopf (2022) cautions, however, that tightness alone is insufficient—it must be paired with rational value orientations to be effective across different pandemic phases. This suggests that the relationship between collectivism and pandemic outcomes is not linear but contingent on complementary cultural traits.

In the GLOBE-only model without demographics, Institutional Collectivism reaches significance ($p = 0.034$), confirming its substantive role. Future research with larger samples could more precisely disentangle the independent contributions of these correlated dimensions.

Figure 4

Bivariate relationship between GLOBE Institutional Collectivism (societal practice) and COVID-19 deaths per 1,000



Institutional Collectivism and COVID-19 Mortality

Note. N = 49. This dimension showed the strongest bivariate correlation but did not reach significance in the multivariate model. (N = 49, r = -0.498, p < 0.001).

4.4 Practices Over Values: What Societies Do Matters More Than What They Believe

One of the most consistent findings in this study is the systematic superiority of GLOBE societal practices over their corresponding values dimensions. Practices outperformed values in seven of nine dimension pairs, often by large margins. The most striking case is Institutional Collectivism: the practice dimension ranked first among all predictors (r = −0.525) while the value dimension showed effectively zero association (r = 0.008).

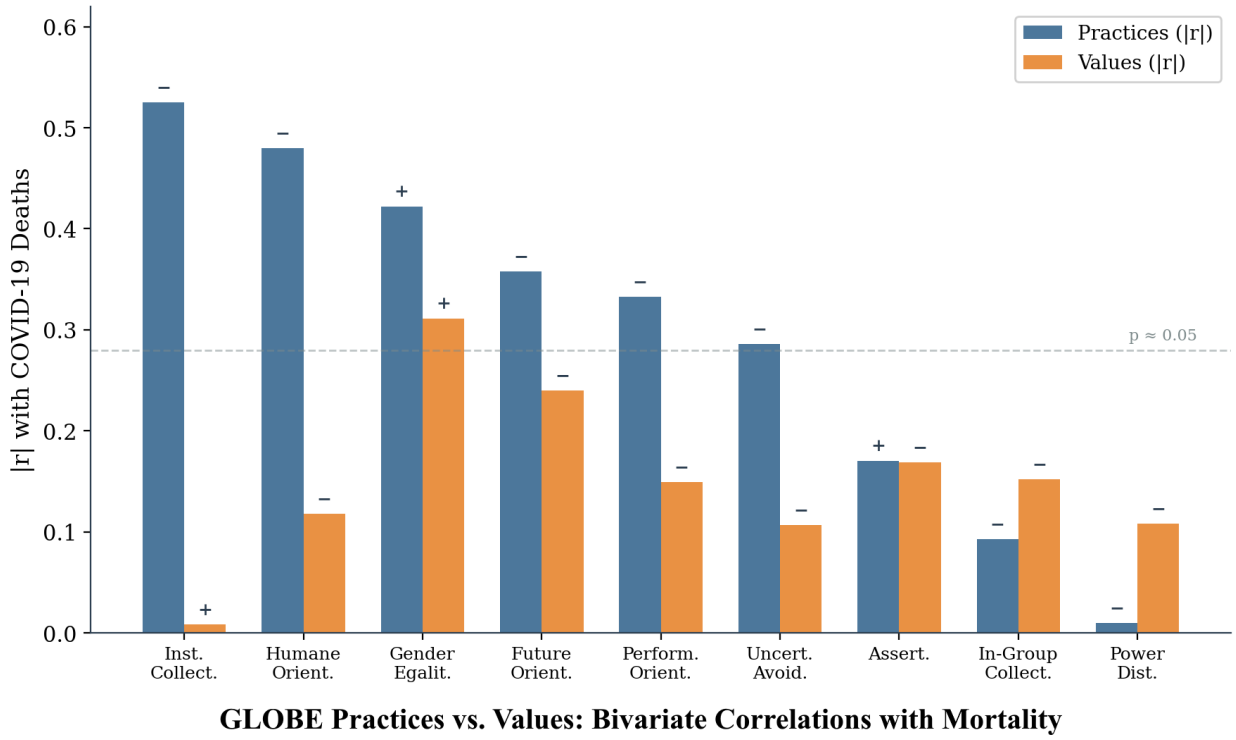
This pattern suggests that what societies actually do—their behavioral norms, institutional routines, and enforcement mechanisms—predicts health outcomes far better than what they aspire to. A society may value collective action in the abstract while lacking the institutional infrastructure to enact it, or may express egalitarian ideals without having reorganized daily life around them. The gap between aspiration and behavior is evidently consequential for pandemic outcomes.

This finding is reinforced by the Hofstede–GLOBE comparison on Uncertainty Avoidance, which provides a particularly instructive case study. Hofstede's Uncertainty Avoidance measures the emotional discomfort people feel about ambiguity—essentially, the feeling of anxiety about the unknown. It correlated positively with mortality ($r = 0.337$), meaning societies that feel more anxious about uncertainty experienced more deaths. Prior research has linked Hofstede's Uncertainty Avoidance to risk aversion and preventive health behavior (de Mooij & Hofstede, 2010) as well as broader health-related behavioral patterns (Katsikitis et al., 2017), establishing it as the most health-relevant of Hofstede's dimensions. GLOBE's Uncertainty Avoidance practice, by contrast, measures the extent to which societies actually deploy rules, procedures, and routines to manage unpredictability. It correlated negatively with mortality ($r = -0.286$), meaning societies that actually built systems to manage uncertainty experienced fewer deaths.

These two constructs correlate with each other at $r = -0.70$ and point in opposite directions relative to pandemic mortality. The divergence is revealing: feeling anxious about uncertainty does not protect against a pandemic; building systematic behavioral responses to uncertainty does. Societies where anxiety about the unknown translates into institutional preparedness fare better than those where the anxiety remains unresolved at the emotional level. This directly supports the broader finding that behavioral practices, not psychological states or aspirational values, drive real-world health outcomes.

Figure 5

Absolute bivariate correlations with COVID-19 mortality for GLOBE practice and value dimensions



Note. Practices outperformed values in seven of nine dimension pairs. Signs (+/−) indicate direction of association.

4.5 Hofstede's Limited Explanatory Power

Despite its widespread use in cross-cultural research, Hofstede's framework performed poorly as a predictor of pandemic outcomes. The Demographics + Hofstede model did not reach overall significance ($p = 0.131$), and adding Hofstede's six dimensions to demographics improved adjusted R^2 by only 0.006. Five of six Hofstede dimensions showed no significant association with mortality; only Uncertainty Avoidance reached significance ($p = 0.048$).

This does not mean Hofstede's framework lacks value for other applications. Taras et al. (2022) have demonstrated that Hofstede's dimensions can predict longevity and health outcomes cross-nationally, suggesting the framework retains utility for health research. However, its more abstract, values-oriented dimensions appear too coarse or misaligned with the behavioral

dynamics that drive pandemic outcomes. The GLOBE framework's behavioral specificity—measuring what societies actually do rather than what they believe—provides a more actionable foundation for public health modeling. Researchers examining culture–health relationships should consider prioritizing practice-based cultural measures over value-based ones.

4.6 Model Validation: The Case of China

China was excluded from the analysis due to concerns about reporting reliability, but the model's treatment of China provides an independent validation of both the exclusion decision and the model's predictive utility.

When China's cultural and demographic profile is entered into the primary model—moderate institutional collectivism, low gender egalitarianism, moderate humane orientation, high population density, median age of 37.4—the model predicts approximately 1,060 deaths per million. China reported 2.8 deaths per million. This represents a divergence of approximately 380:1 between predicted and reported values.

For context, the largest residual among the 49 included countries corresponds to a divergence of approximately 2:1. China's departure from the model is not an outlier in the conventional statistical sense—it is a categorical anomaly that cannot be explained by normal variation, measurement error, or model imprecision.

Critically, this finding is not derived from assumptions about Chinese governance, press freedom, or political transparency. It emerges organically from a statistical model built on cultural and demographic variables that demonstrably predict mortality across 49 countries with varying political systems, reporting standards, and healthcare capacities. The model does not "know" that China's data may be unreliable—it simply identifies that China's reported outcomes are incompatible with what its own cultural and structural profile would predict.

This constitutes independent, quantitative evidence consistent with the assessments of widespread underreporting documented by The Economist's excess mortality model, the World Health Organization, and multiple epidemiological studies. It also demonstrates the model's capacity to identify reporting anomalies—a potentially valuable application for future pandemic surveillance.

4.7 Health Expenditure and Pandemic Outcomes

Health expenditure per capita showed effectively zero association with COVID-19 mortality ($r = 0.018$, $p = 0.864$). This finding challenges the conventional assumption that higher health spending equates to better pandemic outcomes. The United States, which spent more per capita on healthcare than nearly any country in the dataset, experienced one of the higher death rates. Meanwhile, countries spending a fraction of that amount but embedded in strong collectivist or prosocial cultural structures fared substantially better.

The spending result does not exist in isolation. Hospital beds per thousand, GDP per capita, and incarceration rate (a rough proxy for state enforcement capacity) all showed weak or negligible associations with mortality. The entire category of resource and infrastructure variables performed poorly relative to cultural dimensions.

What these findings collectively suggest is that pandemic outcomes are fundamentally a behavioral problem, not a resource problem. A lavishly funded health system embedded in a culture of high individual mobility and low collective compliance can be outperformed by a modestly funded system in a culture of prosocial norms and institutional coordination. This does not mean resources are irrelevant—adequate healthcare infrastructure is clearly necessary. But the data suggest it is not sufficient. The sufficient condition appears to be behavioral, and behavioral capacity is culturally determined.

The policy implication is that pandemic preparedness budgets should include investment in cultural infrastructure—community resilience networks, prosocial communication frameworks, pre-built collective action protocols—alongside material resources. A dollar spent on building community trust and collective response capacity may outperform a dollar spent on ventilator stockpiles if the cultural environment does not support coordinated use of those resources.

5. Policy Implications

The findings of this study carry direct implications for how pandemic preparedness is conceptualized, funded, and executed—particularly in modernized democracies.

5.1 Egalitarian Societies Need Tailored Preparedness

Countries scoring high on Gender Egalitarianism should anticipate wider transmission surfaces during pandemics and incorporate this into their epidemiological modeling. This does not mean treating egalitarianism as a risk factor to be corrected. It means acknowledging that modernized, mobile, dual-participation societies need different containment strategies than societies where large portions of the population are home-based by default. Contact tracing protocols, workplace closure sequencing, childcare contingency planning during lockdowns, and school closure policies all interact differently with egalitarian social structures and must be designed accordingly.

5.2 Prosocial Messaging Over Authoritarian Mandates

The protective effect of Humane Orientation suggests that pandemic messaging in egalitarian societies should lean into prosocial framing rather than top-down mandates. “Protect your community” may outperform “Follow the rules” in cultures that value individual autonomy but also harbor prosocial instincts. There is substantial evidence that prosocial framing increased mask compliance in Western democracies. Public health communication strategies should be culturally calibrated, drawing on the specific motivational structures of the target population rather than assuming universal compliance mechanisms.

5.3 Invest in Cultural Infrastructure

The near-zero association between health spending and pandemic mortality suggests that resource investment without cultural alignment produces poor returns. Pandemic preparedness budgets should include investment in what might be called cultural infrastructure: community resilience networks, pre-established collective action frameworks, prosocial communication campaigns, and institutional mechanisms for rapid coordination. These investments build the

behavioral capacity that determines whether material resources are effectively deployed during crisis.

5.4 Build Temporary Tightening Capacity

Gelfand (2021) has documented how cultural tightness—strong norms and low tolerance for deviance—predicts more effective pandemic containment. Egalitarian societies tend to be culturally loose, prizing individual freedom and resisting centralized constraint. Anicich et al. (2015) found that hierarchical team dynamics predicted better cohesion and survival in high-stakes environments, suggesting that temporary acceptance of steeper coordination hierarchies during emergencies may be a viable mechanism—even in otherwise egalitarian societies. The challenge for these societies is not to permanently tighten their cultural norms—which would undermine the values that make them function well in normal times—but to build pre-negotiated mechanisms for temporary tightening during emergencies. This includes pre-authorized emergency powers with built-in sunset clauses, pre-established social contracts about what collective sacrifices will look like, and institutional protocols that can rapidly shift from loose to tight without requiring ad hoc political negotiation in the middle of a crisis.

5.5 Culturally Calibrated Surveillance

The model's ability to identify China's reporting anomaly suggests a broader application: cultural and demographic profiles can be used to generate expected mortality ranges for countries, and significant deviations from those ranges can flag potential underreporting. This approach could supplement existing surveillance tools by providing an independent, model-based check on reported data quality.

6. Limitations

Several limitations should be noted when interpreting these findings.

First, this is a cross-sectional, correlational analysis. The associations identified do not establish causal relationships between cultural dimensions and COVID-19 mortality. Cultural traits may influence pandemic outcomes through multiple indirect pathways—policy adoption, behavioral compliance, social trust, institutional effectiveness—that this study does not decompose.

Second, the analysis operates at the national level, and associations observed at this aggregate level do not necessarily hold for individuals or subnational groups. This ecological fallacy concern is inherent in all cross-national cultural research. Countries contain substantial internal cultural variation, and the GLOBE and Hofstede scores represent central tendencies that may mask important within-country heterogeneity.

Third, the primary model's sample size (N = 49) is limited by the availability of GLOBE data, which covers 62 societies. While this is adequate for the number of predictors examined (VIF values remain below 6), it constrains statistical power and limits the detection of smaller effects. The non-significance of Institutional Collectivism in the multivariate model, despite its strong bivariate association, may partly reflect this power limitation.

Fourth, COVID-19 mortality data are subject to reporting biases that vary systematically across countries. While the exclusion of China addresses the most extreme case, less dramatic underreporting in other countries—particularly those with limited press freedom or statistical infrastructure—may introduce noise or systematic bias. Future research should test the robustness of these findings using excess mortality estimates where available.

Fifth, the static nature of the analysis does not capture temporal dynamics. Cultural norms may interact differently with pandemic outcomes at different phases—early containment, peak transmission, vaccination rollout—and the December 2021 cutoff captures only a cumulative snapshot. Longitudinal designs could reveal whether cultural predictors shift in strength or direction across pandemic phases. Additionally, cultural traits may interact with risk perception in complex ways (van der Linden, 2015) that aggregate cross-national analysis cannot fully decompose.

Finally, this study examines only two cultural frameworks. The Schwartz cultural value framework (Schwartz, 2006) was considered but excluded because its country coverage reduced sample sizes without adding explanatory clarity beyond what GLOBE practices already provided. Other cultural constructs—including Gelfand's tightness–looseness, Inglehart–Welzel cultural maps, and social trust indices—may offer complementary or competing explanations.

7. Conclusion

This study demonstrates that cultural dimensions—particularly those governing gender dynamics, prosociality, and collective orientation—are not peripheral to pandemic outcomes but central to understanding national variation in COVID-19 mortality. GLOBE societal practices explain over half the cross-national variance when combined with basic demographic controls, dramatically outperforming Hofstede's more abstract value dimensions and rivaling the explanatory power of age structure.

Three findings carry particular weight. First, Gender Egalitarianism emerged as the most robust and consistent predictor of mortality, operating through the structural exposure surfaces that accompany modernized, egalitarian social organization. This finding is not an argument against equality but a recognition that social progress reshapes disease transmission dynamics in ways that require anticipatory planning. Second, the systematic superiority of behavioral practices over aspirational values across seven of nine GLOBE dimensions indicates that pandemic preparedness models should attend to what societies do, not what they believe. Third, the near-zero association between health spending and mortality suggests that the prevailing emphasis on material resources in preparedness frameworks is incomplete without attention to the cultural and behavioral context in which those resources are deployed.

Culture should not be treated as residual variance or noise in epidemiological models. It is a structured and influential system that shapes how societies interpret, implement, and respond to health directives. As Gelfand (2021) notes, cultural evolution does not always keep pace with global threats; even prosocial norms may be poorly calibrated to novel crises, producing mismatches that reduce the effectiveness of otherwise beneficial social structures.

Future pandemic planning should incorporate these insights not merely to predict outcomes but to design culturally calibrated interventions—messaging that resonates with local motivational structures, institutional mechanisms that leverage existing cultural strengths, and preparedness frameworks that account for the specific vulnerability profiles of modernized democracies. The next pandemic will not wait for cultural norms to evolve. The question is whether policy frameworks will be ready to work with the cultures they have.

8. Data Availability Statement

All data used in this study are publicly available from the original sources cited in the manuscript. COVID-19 mortality data were obtained from Our World in Data. Cultural dimension scores were collected from the GLOBE Project publications and Hofstede Insights database. Demographic data were sourced from the United Nations, World Bank, and national statistical agencies. A compiled dataset and replication code will be made available upon publication.

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Appendix A: Robustness Analyses

This appendix presents a series of robustness checks to assess the stability of the primary model's findings. All analyses use the same dataset (N = 49, China excluded) and COVID-19 deaths per 1,000 as the dependent variable unless otherwise noted.

A.1 Reduced Model

To address concerns about the ratio of predictors to observations in the primary 12-predictor model, we estimated a reduced model retaining only the two significant cultural predictors (Gender Egalitarianism and Humane Orientation) alongside the three demographic controls. This yields a predictor-to-observation ratio of approximately 10:1, in line with conventional guidelines.

Table A1

Reduced model: Demographics + Gender Egalitarianism + Humane Orientation

Variable	B	SE	β	t	p	Sig.
(Intercept)	−1.370	1.872	—	−0.73	0.468	
Median Age	0.025	0.019	0.240	1.36	0.180	
Gini Coefficient	0.055	0.018	0.462	3.04	0.004	**
Population Density	<−0.001	<0.001	−0.419	−3.96	<0.001	***
GLOBE Humane Orientation	−0.856	0.212	−0.499	−4.04	<0.001	***
GLOBE Gender Egalitarianism	0.912	0.267	0.377	3.41	0.001	**

Note: N = 49, Adjusted R^2 = 0.514.

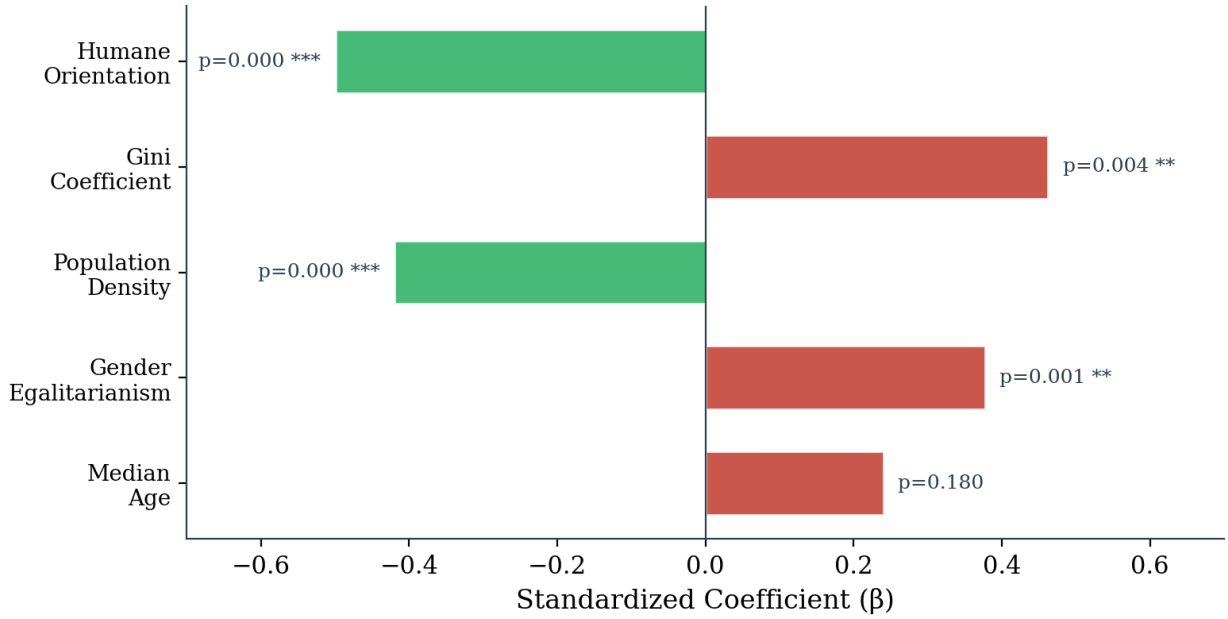
** p < .005. *** p < .001.

The reduced model achieves an adjusted R^2 of 0.514, marginally higher than the full 12-predictor model (0.512), confirming that the additional GLOBE dimensions contribute negligible incremental explanatory power. Critically, both Gender Egalitarianism (p = 0.001) and Humane

Orientation ($p < 0.001$) strengthen in significance compared to the full model, ruling out overfitting as the source of their effects. The Gini coefficient also reaches strong significance ($p = 0.004$), suggesting that income inequality contributes independently to pandemic vulnerability once the noise of redundant predictors is removed.

Figure 6

Standardized coefficients for the reduced model



Standardized Effect Size of Reduced Model

Note. N = 49, Adj. $R^2 = 0.514$. Green indicates protective association; red indicates risk association.

** $p < .005$. *** $p < .001$.

A.2 Influence Diagnostics

Cook's Distance was computed for each observation in the primary model to identify influential data points. Three countries exceeded the conventional threshold of $4/N = 0.082$: Singapore (Cook's D = 11.27), Sweden (0.20), and Egypt (0.10).

Singapore's extreme influence stems from its population density (20,443 persons per square mile), which is approximately seven times higher than the next densest country in the sample. As a city-state, Singapore occupies a unique position in any cross-national analysis, and its high

leverage (0.99) means the model effectively uses Singapore's data to determine its own predicted value. This does not invalidate its inclusion but warrants noting.

To test the stability of key findings, the primary model was re-estimated after removing all three influential observations (N = 46). Results are presented in Table A2.

Table A2

Sensitivity analysis

Variable	Full B	Full p	Trimmed B	Trimmed p	Stable?
Gender Egalitarianism	1.103	0.002	1.115	0.003	Yes
Humane Orientation	−0.801	0.027	−0.724	0.050	Yes
Inst. Collectivism	−0.462	0.224	−0.584	0.131	Yes
Gini Coefficient	0.044	0.041	0.050	0.037	Yes
Population Density	<−0.001	0.007	<−0.001	0.317	Changed
Median Age	0.033	0.130	0.047	0.047	Changed

Note: Primary model with three influential observations removed (N = 46). Adjusted R^2 = 0.557.Singapore (Cook's D = 11.27), Sweden (0.20), and Egypt (0.10).

Gender Egalitarianism remains significant at p = 0.003 with a nearly identical coefficient (1.115 vs. 1.103), confirming that the finding is not driven by a small number of influential countries. Humane Orientation remains significant at p = 0.050. Population density loses significance when Singapore is removed, as expected given Singapore's extreme leverage on this variable. Median age gains significance (p = 0.047), suggesting that Singapore's unusual density was partially masking the age effect. The overall model's adjusted R^2 increases to 0.557, indicating that the influential observations were introducing noise rather than driving the results.

A.3 Distributional Diagnostics

The dependent variable (deaths per 1,000) exhibited mild positive skewness (0.49) and slight platykurtosis (−0.83). A Shapiro-Wilk test on the primary model's residuals yielded W = 0.970,

$p = 0.247$, indicating no significant departure from normality. This confirms that the standard OLS assumptions are met and that significance tests are valid.

A log-transformed specification was also tested. The log model yielded a lower adjusted R^2 (0.418 vs. 0.512), and its residuals failed the Shapiro-Wilk normality test ($W = 0.915$, $p = 0.002$). Gender Egalitarianism remained borderline significant ($p = 0.051$) under log transformation, while Humane Orientation lost significance ($p = 0.116$). However, because the untransformed model better satisfies distributional assumptions and achieves higher explanatory power, it represents the more appropriate specification. The log-transformed results are noted here for completeness but should be interpreted in light of the violated normality assumption in that specification.

A.4 Heteroskedasticity Test

A Breusch-Pagan test was conducted on the primary model's residuals to assess heteroskedasticity. The test statistic was 15.25 ($df = 12$, $p = 0.228$), indicating no significant heteroskedasticity. Standard errors and associated p-values in the primary model are therefore reliable.

A.5 China: Prediction Interval

To supplement the point estimate discussed in Section 4.6, we computed the 95% prediction interval for China based on the primary model. The model predicts 1,060 deaths per million (1.06 per 1,000) for China, with a 95% prediction interval of approximately −540 to 2,659 deaths per million. The lower bound is substantively meaningless (deaths cannot be negative), reflecting the inherent width of prediction intervals in small-sample OLS with 12 predictors.

China's reported value of 2.8 deaths per million technically falls within this wide interval. However, this must be interpreted in context. The prediction interval's breadth reflects model uncertainty, not plausibility of the reported value. The point estimate divergence of 380:1 remains the most informative statistic. Among all 49 included countries, the largest ratio of predicted to actual deaths is approximately 2:1. China's 380:1 ratio represents a qualitatively different category of departure that is not attributable to normal model imprecision.

A.6 Countries Included in Primary Analysis

The following 49 countries had complete data for all demographic and GLOBE practice variables and were included in the primary model: Albania, Argentina, Australia, Austria, Brazil, Canada, Colombia, Costa Rica, Denmark, Ecuador, Egypt, El Salvador, Finland, France, Georgia, Germany, Greece, Guatemala, Hungary, India, Indonesia, Iran, Ireland, Israel, Italy, Japan, Malaysia, Mexico, Morocco, Netherlands, New Zealand, Nigeria, Philippines, Poland, Portugal, Russia, Singapore, Slovenia, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Kingdom, United States, Venezuela, Zambia, and Zimbabwe.